

Mixed Traces PCB Design Competition

Progress Evaluation Submission — 3 September 2026

STM32F401RE Evaluation Board

by Payload Circuits

Adarsh S — Solo Entry

1. Selected MCU and Design Concept

The fixed microcontroller for this competition is the STM32F401RET6 (LQFP64), as required by the brief. The design concept is a compact, self-contained STM32F401RE evaluation board inspired by the NUCLEO-F401RE reference, built to fit within a 100mm × 100mm, two-layer PCB.

Per the competition rules, the on-board ST-LINK debugger is explicitly not required, which simplified the design considerably compared to the Nucleo reference — programming and debugging are handled through a standard external SWD header rather than an on-board programmer IC. This freed up board area and routing complexity that was redirected toward a more generous GPIO expansion header set.

The design philosophy prioritized correctness and routing/documentation quality (the criteria weighted most heavily in evaluation) over chasing extremes in size or cost, and favored “smart” flexibility — a single, well-executed power path and full signal breakout to headers — over redundant ports or power-input options that would add cost and routing complexity without improving the score.

2. Block Diagram

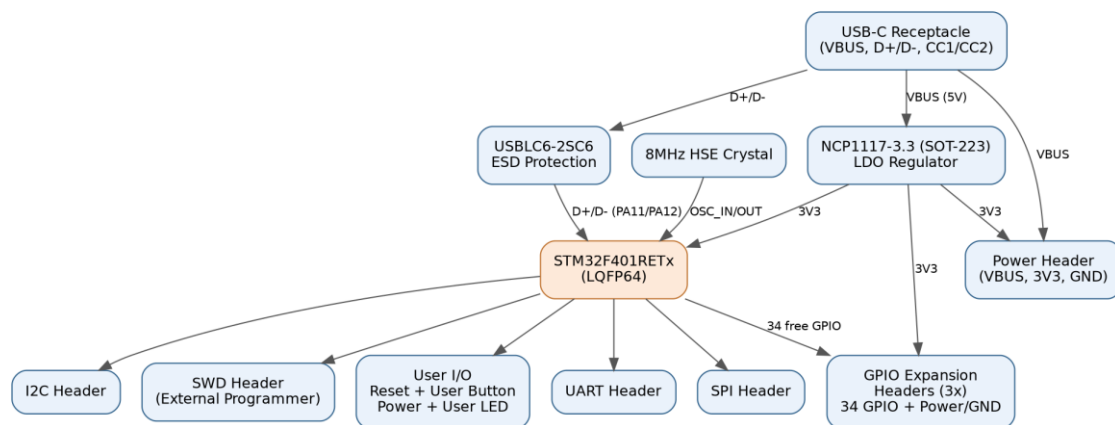


Figure 1: System block diagram — power, USB, MCU, and header breakout

3. Schematic Progress

The schematic is complete and passes ERC with zero unresolved violations.

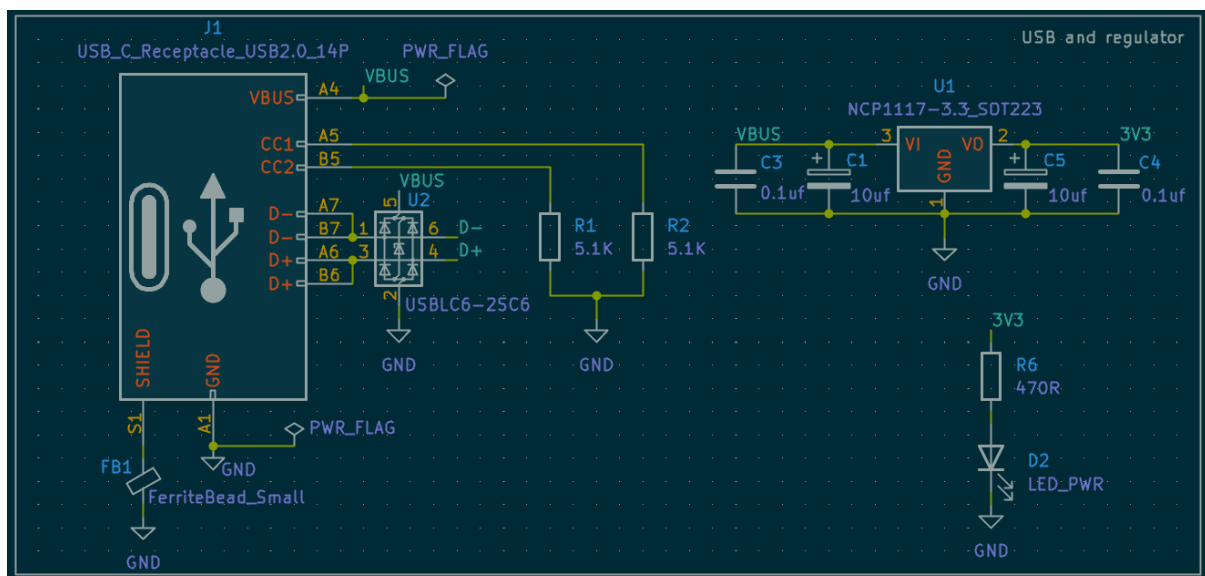


Figure 2: USB-C input, ESD protection, and 3.3V regulation



Figure 3: STM32F401RE core — decoupling, VCAP, reset, BOOT0/DFU, and HSE crystal

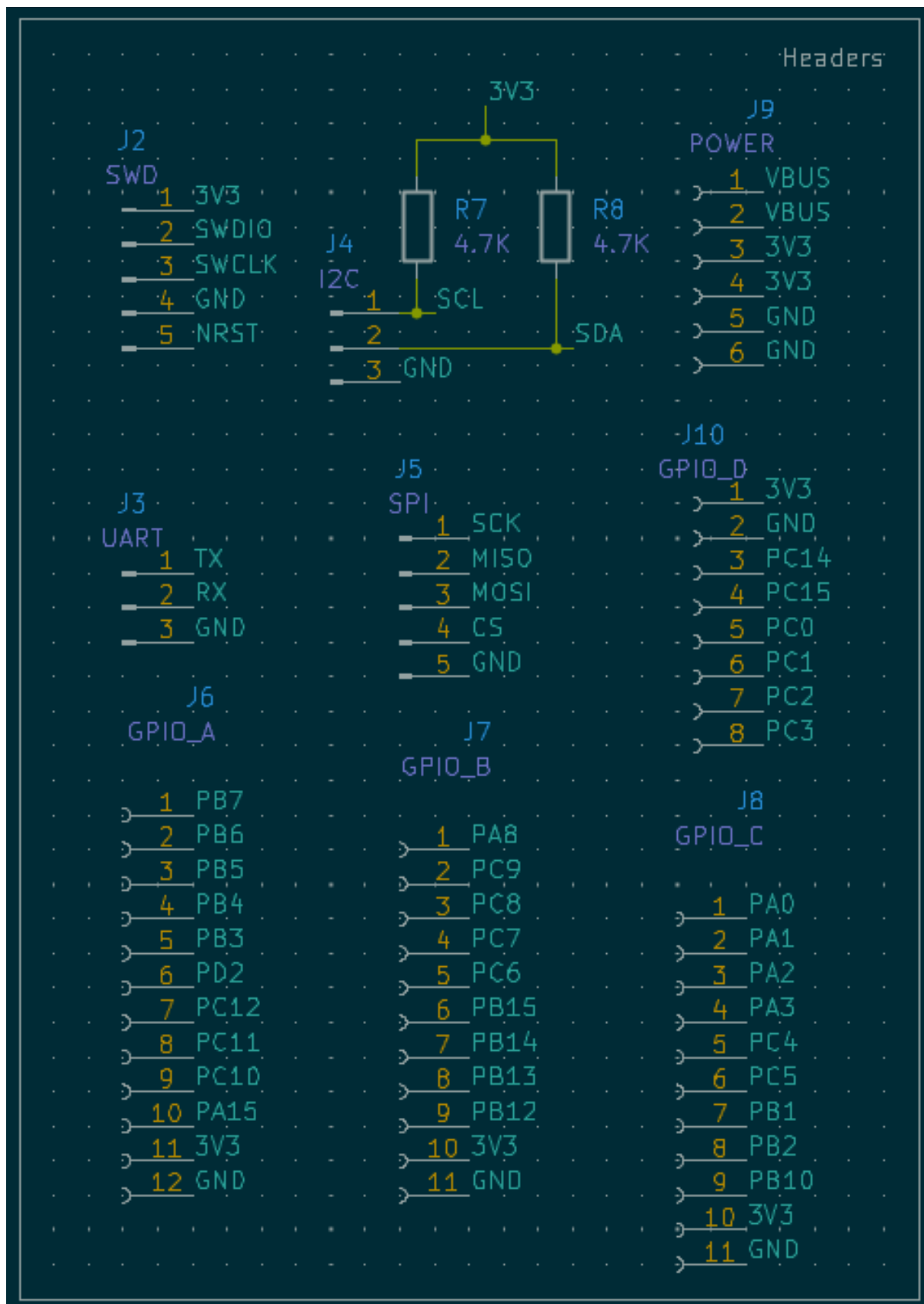


Figure 4: SWD, UART, SPI, I2C, POWER and GPIO expansion header breakout

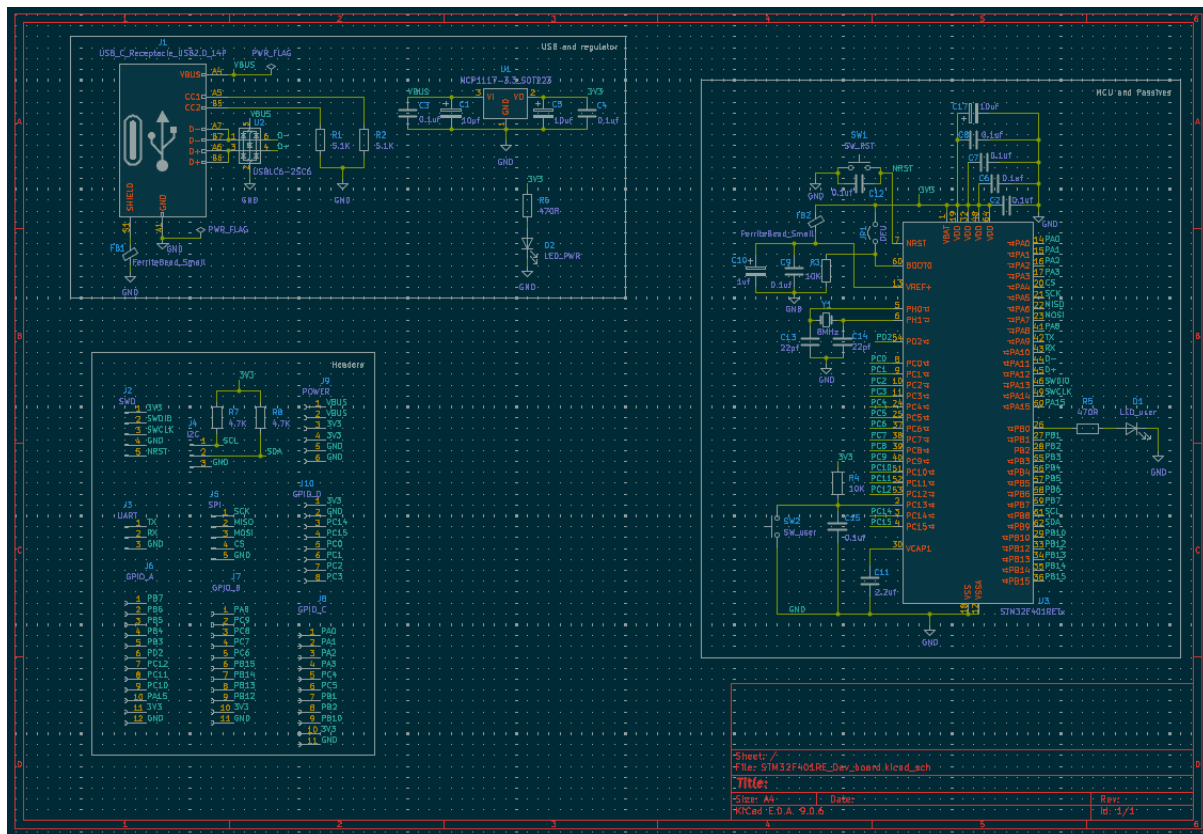


Figure 5: Complete Schematic diagram

4. Component Selection

Key components and the reasoning behind each selection:

Function	Part / Choice	Package	Rationale
MCU	STM32F401RETx	LQFP64	Fixed by competition rules
Voltage regulator	NCP1117-3.3 (ST suffix)	SOT-223	Simple fixed-3.3V LDO, adequate for MCU + peripheral current draw, no external feedback network needed
USB connector	USB-C receptacle, full THT (signal + shield legs)	THT, 2-layer footprint	Mechanically robust for repeated insertion; design-only competition so hand-solder ease was not a constraint
USB ESD protection	USBLC6-2SC6	SOT-23-6	Low-capacitance ESD array for D+/D-lines, standard practice for USB2.0 device ports
HSE crystal	8MHz, 22pF load caps	SMD 3.5x6mm, hand-solder	8MHz cleanly divides/multiplies to both 84MHz SYSCLK and 48MHz USB clock without fractional PLL math
Passives (R, C $\leq 100\text{nF}$)	Generic 0603	0603	Compact, consistent footprint across the board
Bulk/decoupling caps ($\geq 1\mu\text{F}$)	Generic ceramic	0805	Safer margin against DC-bias derating and hand-solder heat vs 0603
Ferrite beads	Generic SMD ferrite bead	0603 (L_0603_1608Metric)	VDD→VDDA analog isolation
Debug/comms/GPIO headers	2.54mm pitch, male (SWD/UART/SPI/I2C) & female (GPIO/Power)	THT 2.54mm	Male for probe/ribbon-cable connections; female for shield/jumper-wire compatibility, matching Nucleo convention

5. PCB Placement / Routing Progress

Component placement is complete, with connectors distributed across all four board edges for accessible wiring, small passives relocated to the bottom layer to relieve top-layer routing density, and the crystal kept immediately adjacent to the MCU's oscillator pins. Routing is fully complete on both layers with a poured ground plane, and the board passes DRC with zero unresolved violations (a small number of connector-specific clearance flags were reviewed and excluded — see Section 6).

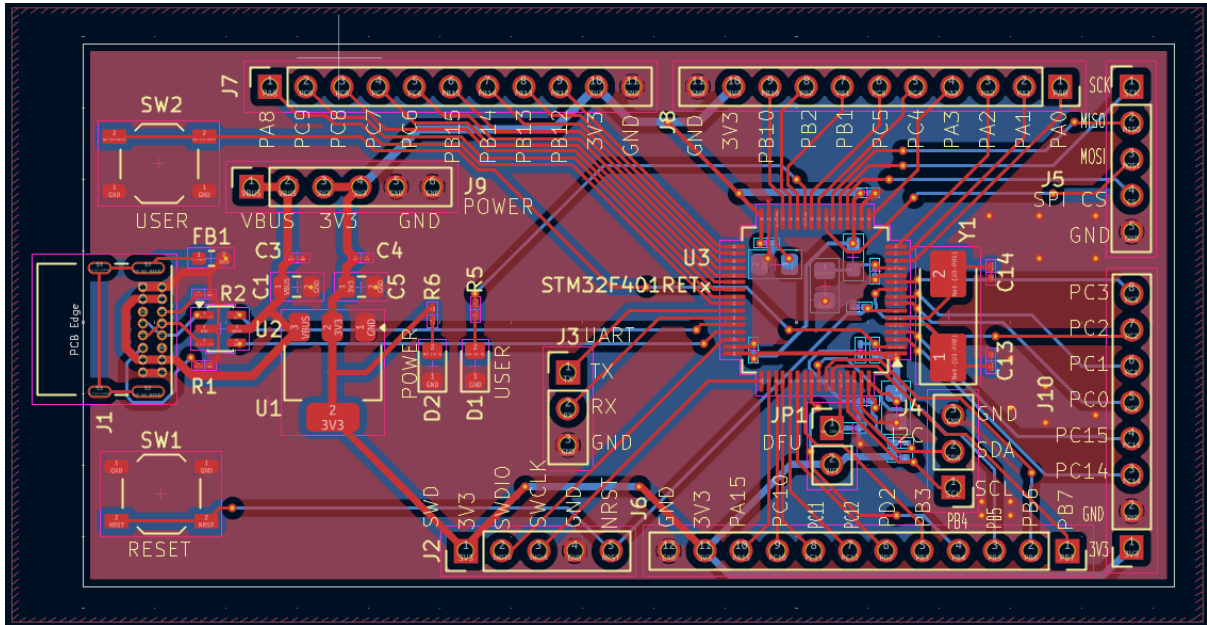


Figure 5: Completed two-layer routing with ground pour (top view, PCB editor)

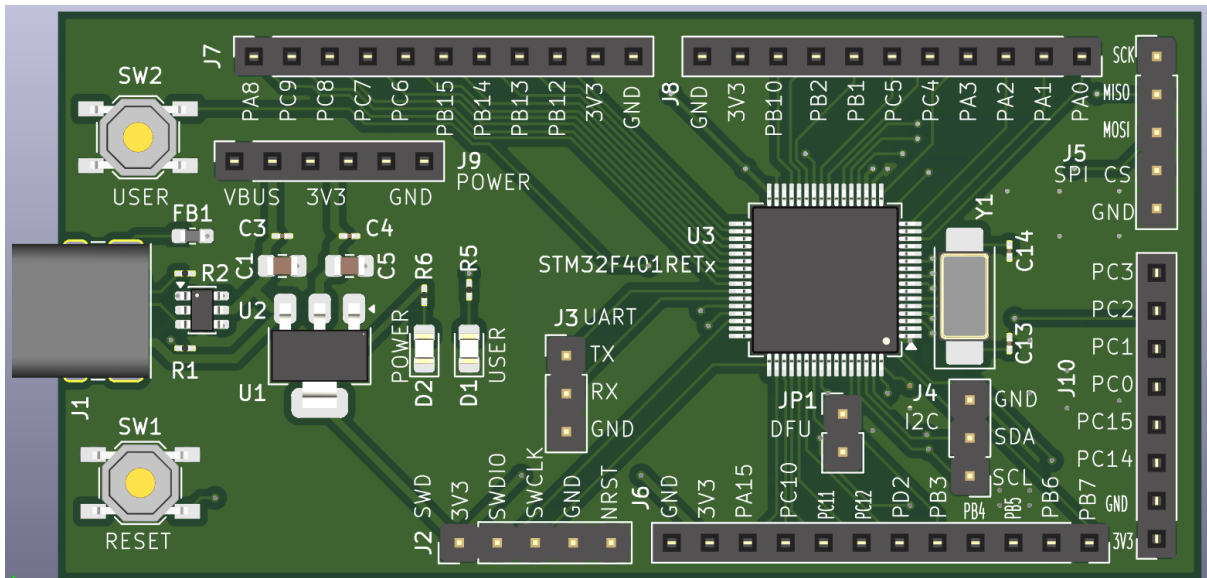


Figure 6: 3D render, top side

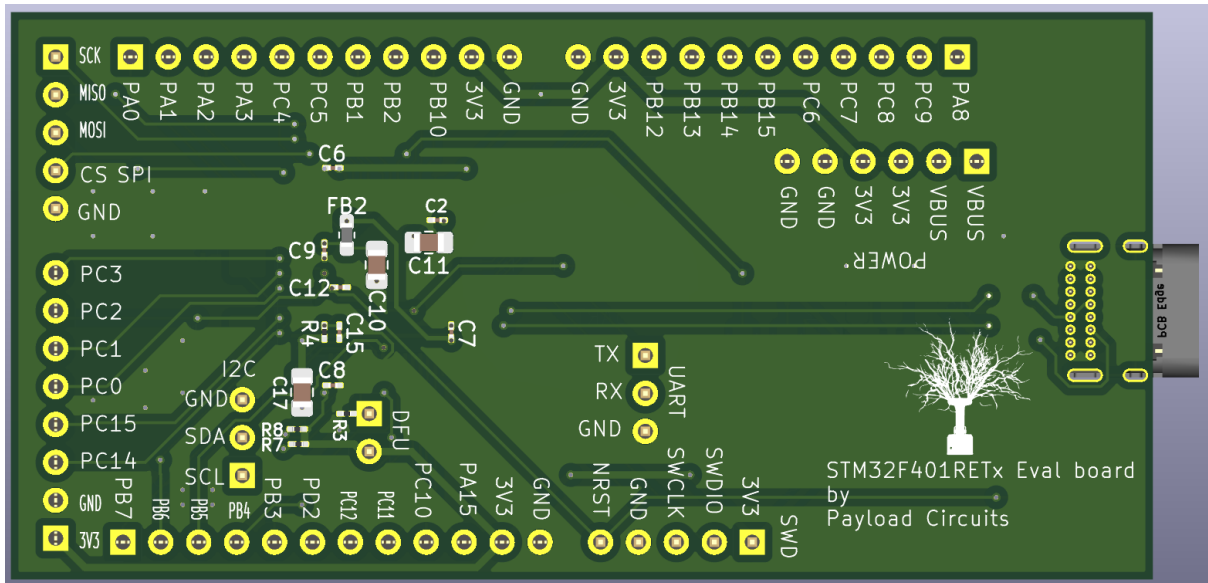


Figure 7: 3D render, bottom side — passives and decoupling network

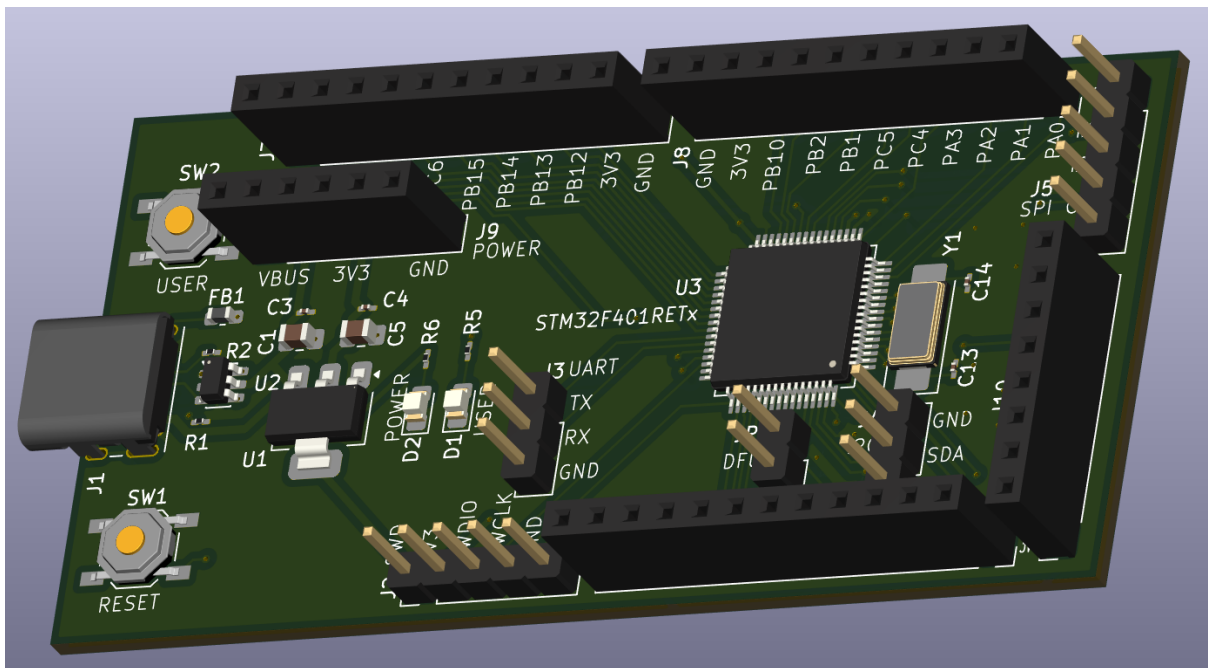


Figure 8: 3D render

6. Major Design Decisions

- On-board ST-LINK omitted per competition rules; programming/debugging handled via an external SWD header instead, saving board area and routing for GPIO expansion.
- USB-C power input (no redundant barrel-jack) with CC1/CC2 pull-downs for USB-C power negotiation, keeping the power section simple, low-cost, and low-routing-complexity.
- NCP1117-3.3 in SOT-223 (not SOT-252/DPAK) selected since MCU + peripheral current draw does not need the larger package's added thermal capacity.
- Full through-hole USB-C receptacle (signal and shield legs) chosen over an SMD-only part for greater mechanical durability under repeated cable insertion.
- 8MHz HSE crystal selected specifically because it cleanly derives both the 84MHz system clock and the 48MHz USB clock without fractional PLL configuration.
- All four MCU VDD pins individually decoupled with 100nF capacitors; a single shared 10μF bulk capacitor covers the common VDD net rather than duplicating bulk capacitance at every pin.
- VDDA/VSSA isolated from the main VDD/VSS supply via a ferrite bead on the VDD→VDDA path only, with VSS→VSSA tied directly, following standard STM32 analog-supply practice.
- GPIO expansion split across four headers by port, with 3V3/GND reference pins added to every header — directly satisfying the brief's "power and ground connections through headers" requirement rather than treating it as GPIO-only.
- Small passives (resistors, decoupling/bulk caps, ferrite beads) moved to the bottom copper layer during layout to relieve routing congestion around the MCU on the top layer, while mechanically-interactive parts (connectors, buttons, LEDs, crystal, MCU) remained on top.
- A small number of DRC clearance/thermal-relief flags on the USB-C connector's own footprint (internal pin spacing, F.Fab/Edge.Cuts overlap) were individually reviewed and excluded as expected connector-footprint behavior rather than genuine manufacturing risk; all other violations were resolved by repositioning or adjusting zone connection settings.